

Parthapratim Chatterjee

Third Year Undergraduate, IIT Kanpur
Department of Material Science and Engineering

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Academic Qualifications

Year	Educational Qualification	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology Kanpur	7.60/10
2021	CBSE(XII)	Mansukhbhai Kothari National School, Pune	95.2%
2019	ICSE(X)	Vibgyor High International School NIBM, Pune	95.4%

Scholastic Achievements

- Secured **All India Rank 5980** in **Joint Entrance Examination Advanced 2021** among the 1.5 Lakh shortlisted candidates.
- Secured **All India Rank 6858** in **Joint Entrance Examination Mains 2021** among the 1 Million shortlisted candidates.

Work Experience

Generation of 3D objects from 2D image| Internship | *Motherson Technology Services Limited, Noida* December'23

Objective	• To generate a 3D CAD compatible design from 2D sketch and given desired design parameters.
Approach	• Designed a Generative AI network to train using available design datasets to generate 3D objects from 2D images. • AI-generated 3D objects are compatible with CAD design software for further editing and accuracy. improvement.
Result	• Generated a AI engine that could generate 3D design from 2D input which is compatible with CAD softwares.

Synthetic Microstructure Data Generation For Machine Learning Composite Architecture Prediction | **SURGE'23**
Prof. Shikhar Krishn Jha May'23-July'23

Objective	• To predict properties of materials computationally as an alternative to physical experimentation using dataset.
Approach	• Designed an algorithm in Python to generate Synthetic Microstructure images to form a dataset for the input. • Automated the FEniCS code to generate the Material Property plots for the dataset of microstructure images. • Trained GAN model for the prediction of property plots from input image using Image to Image translation.
Result	• Generated a diverse dataset of images of Microstructures for input to GAN model for training and prediction. • Increasing iterations led to highly accurate model-generated material plots, confirmed by minimal RMS error .

Key Projects

- Crane, Manufacturing Processes 1, TA-201** | *Course Project* | *Prof. Nilesh Badwe* January'23-May'23
 - Prepared a **CAD design** using **Fusion 360** of a crane model as a part of this course with **7** other members of the group.
 - Conducted material analysis for optimal project efficiency, ensuring **accurate selection** and allocation of resources for completion.
 - Explored various Manufacturing Processes like **Drilling, Casting, Sheet Metal Forming, Welding, Brazing** and **Molding**.
- Computer Vision and Sensors of AUTOMI 🤖** | *Team Humanoid, IIT Kanpur* | *Prof. K S Venkatesh* December'22
 - Trained a custom **YOLOv5** model to accurately detect **Ping-Pong balls** using a custom dataset with **augmented images**.
 - Implemented method to **detect center** of circle of revolving archery board and **trajectory** that was followed was stored.
 - Developed code to **track lines** and **change direction** based on detected arrows, such as turning left when detecting a left arrow.
- Automated Door Mechanism, Manufacturing Processes 2, TA-202**| *Course Project* | *Prof. Arvind Kumar* Aug'22-Nov'22
 - Prepared a **CAD design** using **Inventor** of Automatic Door Mechanism as a part of this course with **8** other members of the group.
 - Intergrated a light sensor with **Arduino** in such a way that the door would open when light was detected by it using **1** motor.
 - Explored about the various Manufacturing Processes like **Drilling** and **Milling** and majorly used **Mild Steel** for the project.
- Regression Analysis 📊**| *Stamatics Society, IIT Kanpur* May'22-July'22
 - Implemented ML algorithms such as **Linear, Logistic** and **Polynomial Regression** with **Cost function** and **Gradient Descent**.
 - Implemented a **Logistic Regression** model from scratch to predict diabetes from a dataset using BMI, Age and other factors.
 - Constructed a **Multivariate Regression model** with **R2 score of 0.76** to **accurately predict** Insurance price from **5** factors.

Technical Skills

Programming Languages	Operating Systems and Softwares	Tools and Libraries
C, C++, Python, Java, MySQL, MATLAB, L ^A T _E X, HTML	Windows, Ubuntu, Autodesk Fusion 360, MacOS, Microsoft Office	Numpy, Pandas, Matplotlib, Tensorflow, Keras, OpenCV, SciPy, Git, Scikit-Learn

Positions of Responsibility

- DUGC Students' Nominee, Materials Science and Engineering** | **Students Gymkhana, IIT Kanpur** Sept'23- Present
 - Collaborated with **professors** of the department for handling any problems faced by students in their academic programmes.
- Manager, Research and Projects** | **Materials Society, IIT Kanpur** Jul'23- Present
 - Organised various **Academic Oriented** events like **Alumni talks, Fresher Orientation, Farewell** and **Intern Sessions**.
- Team Member** | **Team Humanoid, IIT Kanpur** Aug'22-May'23
 - Helped to organise events like **SnT Pavilion** where the team's past work was shown in front of **1000+** students over three days.

Relevant Coursework

Fundamentals of Computing (A)	Introduction to Electronics	Quantum Physics
Nature and Properties of Materials (A)	Rate Processes	Introduction to Biomaterials
Thermodynamics and Phase Equilibria	Structure and Characterization of Materials	Mechanics of Solids

Extra-Curricular Activities

- Secured **First** position in **Inter House Cricket Competition** and **First** position in **Presentation of a Speech** in School.
- Participated in **3rd Pauling Memorial Inter-School Science Quiz** organized by **IISER, Pune** on **National Science Day**.
- Coordinated in conducting **Technocruise** which is a pre zonal round of **Techkriti** as a **Senior Executive** in **Techkriti**.
- Completed the **MUNtor Program** and received the beginner certification from **MUN Cafe** for excelling in debate and diplomacy.